

```
Python 3.14.4 (tags/v3.14.4:23116f9, Apr 7 2026, 14:10:54) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Calculate Momentum.py =====
Enter mass in kg: 5
Enter velocity in m/s: 8
Momentum = 40.0 kg m/s
>>>
```

```
Calculate Momentum.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE)...
File Edit Format Run Options Window Help
# Accept mass and velocity
m = float(input("Enter mass in kg: "))
v = float(input("Enter velocity in m/s: "))

# Calculate momentum
momentum = m * v

# Display result
print("Momentum =", momentum, "kg m/s")

Ln: 1 Col: 0
```

Ln: 8 Col: 0

```
Birth Date to Age and Salary Rupees to Dollars.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Birth Date to Age and Salary Rupees to Dollars.py (3.14.4)
File Edit Format Run Options Window Help

from datetime import date

# Input birth year
birth_year = int(input("Enter birth year: "))

# Calculate age
current_year = date.today().year
age = current_year - birth_year

print("Age =", age)

# Salary conversion
salary_rupees = float(input("Enter salary in rupees: "))

# Example conversion rate
dollar_rate = 83

salary_dollars = salary_rupees / dollar_rate

print("Salary in Dollars =", salary_dollars)
```

```
IDLE Shell 3.14.4
File Edit Shell Debug Options Window Help

Python 3.14.4 (tags/v3.14.4:23116f9, Apr 7 2026, 14:10:54) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Birth Date to Age and Salary Rupees to Dollars.py
Enter birth year: 2000
Age = 26
Enter salary in rupees: 60000
Salary in Dollars = 722.8915662650602
>>>
```

Ln 21 Col:0

30°C Clear Search 22:41 09-05-2026

Birth Date to Age and Salary Rupees to Dollars.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Birth Date to Age and Salary Rupees to Dollars.py (3.14.4)

File Edit Format Run Options Window Help

```
from datetime import datetime
# Input
birth_y = int(input("Enter birth year: "))
# Calculations
current_age = datetime.now().year - birth_y
age = current_age
print(f"Your current age is {age} years.")
# Salary
salary_rupees = int(input("Enter salary in rupees: "))
salary_dollars = salary_rupees / 72.89
print(f"Your salary in dollars is {salary_dollars:.2f}.")
# Example
for i in range(10):
    print(f"Year {i+1}: Salary {i*10000} rupees, {i*10000/72.89} dollars")
```

Ln: 1 Col: 0

IDLE Shell 3.14.4

File Edit Shell Debug Options Window Help

Python 3.14.4 (tags/v3.14.4:23116f9, Apr 7 2026, 14:10:54) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Birth Date to Age and Salary Rupees to Dollars.py
Enter birth year: 2000
Age = 26
Enter salary in rupees: 60000
Salary in Dollars = 722.8915662650602
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py
Enter number of terms: 5
Fibonacci Series:
0 1 1 2 3
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py
Enter number of terms: 50
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 24157817 39088169 63245986
>>>

Ln: 18 Col: 210

```
Fibonacci Series Using Recursion.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py (3.14.4)
File Edit Format Run Options Window Help

def fibonacci(n):
    if n <= 1:
        return n
    else:
        return fibonacci(n-1) + fibonacci(n-2)

terms = int(input("Enter number of terms: "))
print("Fibonacci Series:")
for i in range(1, terms + 1):
    print(fibonacci(i), end=" ")
    if i % 5 == 0:
        print()

Number Pattern.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Number Pattern.py (3.14.4)
File Edit Format Run Options Window Help

rows = int(input("Enter number of rows: "))
for i in range(1, rows + 1):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()

IDLE Shell 3.14.4
File Edit Shell Debug Options Window Help

Python 3.14.4 (tags/v3.14.4:23116f9, Apr 7 2026, 14:10:54) [MSC v.1914 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py
Enter birth year: 2000
Age = 26
Enter salary in rupees: 60000
Salary in Dollars = 722.8915662650602

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py
Enter number of terms: 5
Fibonacci Series:
0 1 1 2 3

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py
Enter number of terms: 50
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711
26657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 24157817 39088169 63245986 102334155 165580141 267914296

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Number Pattern.py
Enter number of rows: 2
1
1 2

>>>
```

Windows taskbar showing the Start button, Search bar, and various application icons (including File Explorer, Edge, and the Settings app). The system tray on the right displays the date and time as 09-05-2026, 22:45.

```
Fibonacci Series Using Recursion.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py (3.14.4)
File Edit Format Run Options Window Help

def fibonacci(n):
    if n <= 1:
        return n
    else:
        return fibonacci(n-1) + fibonacci(n-2)

terms = int(input("Enter the number of terms: "))
print("Fibonacci Series:")
for i in range(terms):
    print(fibonacci(i))

Number Pattern.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Number Pattern.py (3.14.4)
File Edit Format Run Options Window Help

# Input marks of 5 subjects
total = 0
pass_status = True

for i in range(1, 6):
    marks = int(input(f"Enter marks of subject {i}: "))
    if marks < 40:
        pass_status = False
    total += marks

percentage = total / 5
print("Percentage = ", percentage)

if pass_status:
    print("Student Passed")
    if percentage > 75:
        print("Grade: Distinction")
    elif percentage >= 60:
        print("Grade: First Division")
    elif percentage >= 50:
        print("Grade: Second Division")
    elif percentage >= 40:
        print("Grade: Third Division")
else:
    print("Student Failed")

IDLE Shell 3.14.4
File Edit Shell Debug Options Window Help

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711
28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 57
02887 9227465 14930352 24157817 39088169 63245986 102334155 165500141 267914296
- RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Number Pattern.py
Enter number of rows: 2
1
1 2
>>>
= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Patterns.py
Enter number of rows: 6
*
**
***
****
*****
>>>
= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Result and Grade.py
Enter marks of subject 1: 20
Enter marks of subject 2: 30
Enter marks of subject 3: 50
Enter marks of subject 4: 80
Enter marks of subject 5: 90
Percentage = 54.0
Student Failed
>>>
= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Result and Grade.py
Enter marks of subject 1: 50
Enter marks of subject 2: 40
Enter marks of subject 3: 60
Enter marks of subject 4: 80
Enter marks of subject 5: 90
Percentage = 64.0
Student Passed
Grade: First Division
>>>
```


Fibonacci Series Using Recursion.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py (3.14.4)

File Edit Format Run Options Window Help

```
def fibonacci(n):
    if n <= 1:
        return n
    else:
        return fibonacci(n-1) + fibonacci(n-2)

terms = int(input("Enter number of terms: "))
print("Fibonacci Series:")
for i in range(1, terms+1):
    print(fibonacci(i))
```

Ln 1 Col: 19

Patterns.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Patterns.py

File Edit Format Run Options Window Help

```
rows = int(input("Enter number of rows: "))
for i in range(1, rows+1):
    for j in range(1, i+1):
        print(j, end=" ")
    print()
```

Ln 1 Col: 19

Reverse Number.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py

File Edit Format Run Options Window Help

```
num = int(input("Enter a number: "))

reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

print("Reverse Number =", reverse)
```

Ln 1 Col: 19

IDLE Shell 3.14.4

File Edit Shell Debug Options Window Help

```
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Patterns.py
Enter number of rows: 6
*
**
***
****
*****
*****

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Result and Grade.py
Enter marks of subject 1: 20
Enter marks of subject 2: 30
Enter marks of subject 3: 50
Enter marks of subject 4: 80
Enter marks of subject 5: 90
Percentage = 54.0
Student Failed

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Result and Grade.py
Enter marks of subject 1: 50
Enter marks of subject 2: 40
Enter marks of subject 3: 60
Enter marks of subject 4: 80
Enter marks of subject 5: 90
Percentage = 64.0
Student Passed
Grade: First Division

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py
Enter a number: 6
Reverse Number = 6

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py
Enter a number: 50
Reverse Number = 5

>>>
```

Ln: 59 Col: 0

Fibonacci Series Using Recursion.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py (3.14.4)

Reverse Star Pattern.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Star Pattern.py (3.14.4)

File Edit Format Run Options Window Help

rows = int(input("Enter number of rows: "))

for i in range(rows, 0, -1):

print(" " * i)

Ln: 1 Col: 0

IDLE Shell 3.14.4

File Edit Shell Debug Options Window Help

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Result and Grade.py

Enter marks of subject 1: 20

Enter marks of subject 2: 30

Enter marks of subject 3: 50

Enter marks of subject 4: 80

Enter marks of subject 5: 90

Percentage = 54.0

Student Failed

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Result and Grade.py

Enter marks of subject 1: 50

Enter marks of subject 2: 40

Enter marks of subject 3: 60

Enter marks of subject 4: 80

Enter marks of subject 5: 90

Percentage = 64.0

Student Passed

Grade: First Division

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py

Enter a number: 6

Reverse Number = 6

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py

Enter a number: 50

Reverse Number = 5

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Star Pattern.py

Enter number of rows: 5

**

*

>>>

Ln: 67 Col: 0

30°C Clear

Search

22:48 09-05-2026


```
Fibonacci Series Using Recursion.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Fibonacci Series Using Recursion.py (3.14.4)
Reverse Star Pattern.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Star Pattern.py (3.14.4)
Single Digit -> Square, Two Digit -> Square Root, Three Digit -> Cube Root.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit -> Square, Two Digit -> Square Root, Three Digit -> Cube Root.py (3.14.4)
File Edit Format Run Options Window Help
import math
n = int(input("Enter a number: "))
digits = len(str(abs(n)))
if digits == 1:
    print("Square =", n ** 2)
elif digits == 2:
    print("Square Root =", math.sqrt(n))
elif digits == 3:
    print("Cube Root =", n ** (1/3))
else:
    print("Number is not 1, 2, or 3 digit")
Ln: 1 Col: 0

IDLE Shell 3.14.4
File Edit Shell Debug Options Window Help
Enter marks of subject 1: 50
Enter marks of subject 2: 40
Enter marks of subject 3: 60
Enter marks of subject 4: 80
Enter marks of subject 5: 90
Percentage = 64.0
Student Passed
Grade: First Division
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py
Enter a number: 6
Reverse Number = 6
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Number.py
Enter a number: 50
Reverse Number = 5
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Reverse Star Pattern.py
Enter number of rows: 5
*****
****
***
**
*
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit -> Square, Two Digit -> Square Root, Three Digit -> Cube Root.py
Enter a number: 55
Square Root = 7.416198487095663
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit -> Square, Two Digit -> Square Root, Three Digit -> Cube Root.py
Enter a number: 222
Cube Root = 6.055048946511104
>>>
Ln: 75 Col: 0
```


MITAOE Home Dashboard My courses

Python Basics.py - C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

```
File Edit Format Run Options Window Help
# Dictionary with Player Name: Height (in cm)
players = {
    "Virat": 175,
    "Rohit": 173,
    "Hardik": 183,
    "Bumrah": 178,
    "Rahul": 180
}

# Tallest person becomes captain
captain = max(players, key=players.get)
print(f"The Captain of the team is: {captain} (Height: {players[captain]}cm)")
```

Ln 13 Col: 0

Opened: Saturday, 18 April 2026 5:31 AM Closes: Saturday, 30 May 2026 5:31 AM

30°C Clear

22:58 09-05-2026

ENG IN

IDE Shell 3.14.4

File Edit Shell Debug Options Window Help

```
>>> Enter a number: 55
Square Root = 7.416198487095663

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit -> Square, Two Digit -> S
quare Root, Three Digit -> Cube Root.py
Enter a number: 222
Cube Root = 6.055048946511104

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit -> Square, Two Digit -> S
quare Root, Three Digit -> Cube Root.py
Enter a number:
Enter number of employees: 55000
Enter salary of employee 1: 30000
Enter salary of employee 2: 85000
Enter salary of employee 3: 60000
Enter salary of employee 4:
= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\statistical analysis.py
Enter salary of employee 1: 20000
Enter salary of employee 2: 50000
Enter salary of employee 3: 60000
Sum = 130000.0
Average = 43333.333333333336
Maximum = 60000.0
Minimum = 20000.0

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Python Basics.py
Enter number to search: 40
Number found at index 3

>>>

= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\cricket.py
The Captain of the team is: Hardik (Height: 183cm)

>>>
```

Ln: 100 Col: 0

MITAOE Home Dashboard My courses

Not secure moodle.mitaoe.ac.in/course/view.php?id=6094

Python Basics.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

cricket.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

Student Marks tuples.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

```
# Marks in a list
marks = [85, 90, 78, 92, 88]
total = sum(marks)
percentage = total / len(marks)

print(f"Total Marks: {total}, Percentage: {percentage}%")

# Tuple Operations
my_tuple = (1, 2, 3, 2, 4)
print("Tuple Count of 2:", my_tuple.count(2))
print("Index of 3:", my_tuple.index(3))
```

File Edit Format Run Options Window Help

File Edit Shell Debug Options Window Help

```
Enter a number: 222
Cube Root = 6.055048946511104

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Single Digit -> Square, Two Digit -> Square Root, Three Digit -> Cube Root.py
Enter a number:
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/statistical analysis.py
Enter number of employees: 55000
Enter salary of employee 1: 30000
Enter salary of employee 2: 85000
Enter salary of employee 3: 60000
Enter salary of employee 4:
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/statistical analysis.py
Enter number of employees: 3
Enter salary of employee 1: 20000
Enter salary of employee 2: 50000
Enter salary of employee 3: 60000
Sum = 130000.0
Average = 43333.333333333336
Maximum = 60000.0
Minimum = 20000.0

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Python Basics.py
Enter number to search: 40
Number found at index 3

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/cricket.py
The Captain of the team is: Hardik (Height: 183cm)

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Student Marks tuples.py
Total Marks: 433, Percentage: 86.6%
Tuple Count of 2: 2
Index of 3: 2
```

Ln: 105 Col: 0

5:31 AM

30°C Clear

23:00 09-05-2026


```
untitled (3.14.4)
practical_2.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune...
File Edit Format Run Options Window Help
# --- PRACTICE LAB ASSIGNMENT ---
# List Operations
my_list = [10, 20, 30]
my_list.append(40)
my_list.remove(20)
print("List Operations Result:", my_list)

# Dictionary Operations
my_dict = {"name": "Ali", "age": 20}
my_dict["city"] = "Mumbai"
print("Dictionary Operations Result:", my_dict)

# --- LAB ASSIGNMENT ---
# 1. Linear Search
def linear_search(arr, target):
    for i in range(len(arr)):
        if arr[i] == target: return i
    return -1

nums = [12, 45, 7, 23, 56]
target = int(input("Enter number to search: "))
print(f"Position: {linear_search(nums, target)}")

# 2. Cricket Team (Tallest as Captain)
team = {"Virat": 175, "Rohit": 173, "Hardik": 183, "Rahul": 180, "Bumrah": 178}
captain = max(team, key=team.get)
print(f"Captain: {captain} (Height: {team[captain]}cm)")

# --- SELF STUDY ---
# 1. Marks and Percentage
marks = [80, 85, 90, 75, 95]
total = sum(marks)
print(f"Total: {total}, Percentage: {(total/len(marks))}%")

# 2. Tuple Operations
my_tuple = (1, 2, 3, 2)
print("Tuple Count of 2:", my_tuple.count(2))

Ln: 38 Col: 0

IDLE Shell 3.14.4
File Edit Shell Debug Options Window Help
Enter salary of employee 2: 85000
Enter salary of employee 3: 60000
Enter salary of employee 4:
- RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/statistical analysis.py
Enter number of employees: 3
Enter salary of employee 1: 20000
Enter salary of employee 2: 50000
Enter salary of employee 3: 60000
Sum = 130000.0
Average = 43333.333333333336
Maximum = 60000.0
Minimum = 20000.0

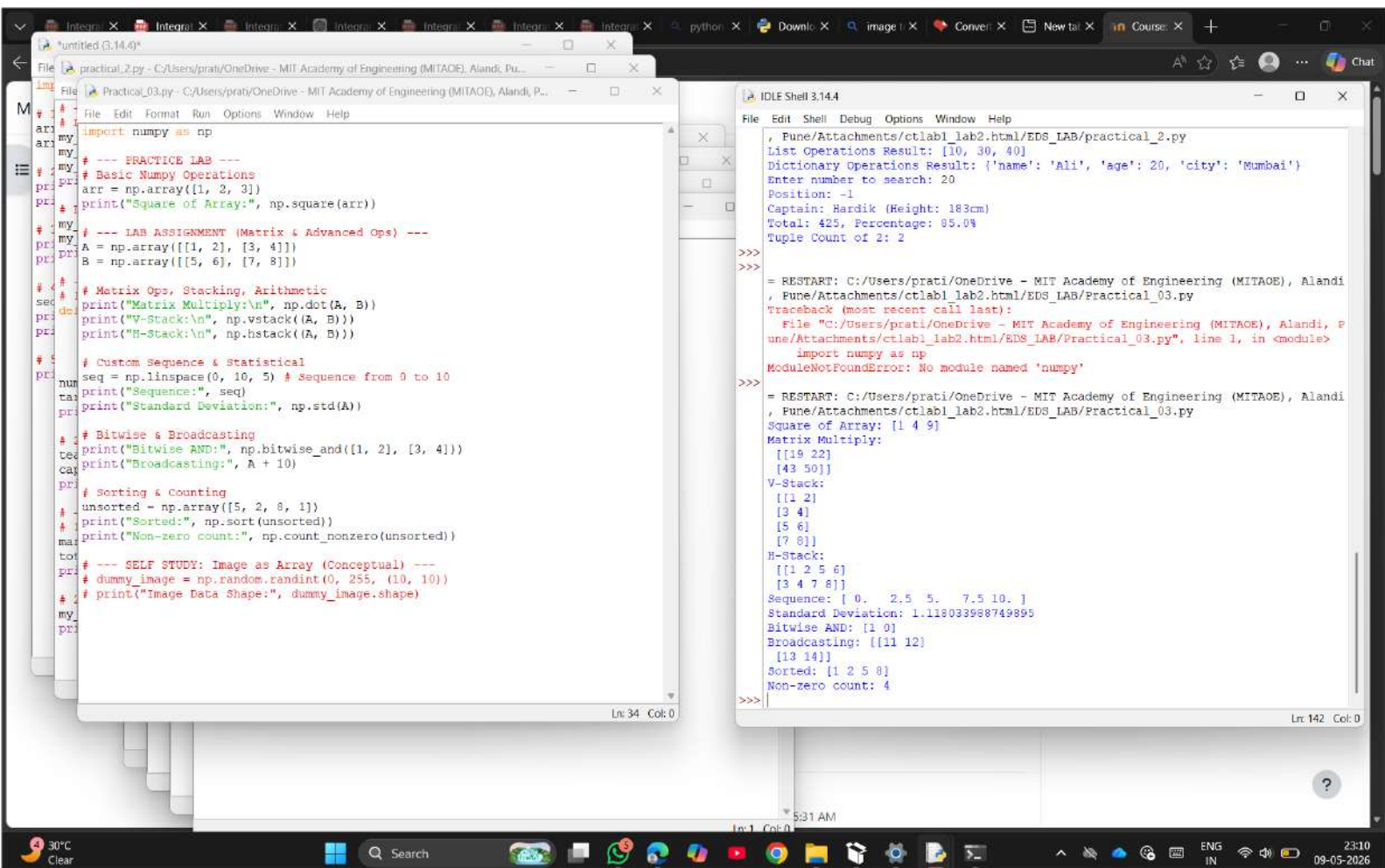
>>>
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Python Basics.py
Enter number to search: 40
Number found at index 3

>>>
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/cricket.py
The Captain of the team is: Hardik (Height: 183cm)

>>>
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Student Marks tuples.py
Total Marks: 433, Percentage: 86.6%
Tuple Count of 2: 2
Index of 3: 2

>>>
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/practical_2.py
List Operations Result: [10, 30, 40]
Dictionary Operations Result: {'name': 'Ali', 'age': 20, 'city': 'Mumbai'}
Enter number to search: 20
Position: -1
Captain: Hardik (Height: 183cm)
Total: 425, Percentage: 85.0%
Tuple Count of 2: 2

>>>
Ln: 115 Col: 0
```




```
untitled (3.14.4)
practical_2.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pu...
practical_03.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, P...
practical_04.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, P...
File Edit Format Run Options Window Help
import pandas as pd
arr =
print(
# --- 1. PRACTICE LAB ASSIGNMENT ---
# All Pandas Operations
s = pd.Series([10, 20, 30])
print("Pandas Series:\n", s)
# --- 2. LAB ASSIGNMENT (Sales Analysis) ---
# Dataset for real life application
data = {
    'Month': ['Jan', 'Feb', 'March', 'Jan', 'Feb', 'March', 'Jan', 'April', 'May',
    'Product': ['Laptop', 'Mouse', 'Laptop', 'Phone', 'Laptop', 'Phone', 'Mouse',
    'City': ['Mumbai', 'Delhi', 'Mumbai', 'Pune', 'Mumbai', 'Pune', 'Delhi', 'Pu',
    'Sales': [50000, 1200, 50000, 20000, 50000, 20000, 1200, 50000, 20000, 50000],
    'Quantity': [1, 2, 1, 1, 1, 1, 2, 1, 1, 1]
}
df = pd.DataFrame(data)
print(df)
# Bitw # 10 Grains (Rows)
print(print("Displaying 10 Grains of Data:\n", df.head(10)))
# Questions
print("\nBest Month for Sales:", df.groupby('Month')['Sales'].sum().idxmax())
print("Product Sold Most (Quantity):", df.groupby('Product')['Quantity'].sum().i
print("City with Most Products:", df.groupby('City')['Quantity'].sum().idxmax())
# --- 3. SELF STUDY ASSIGNMENT ---
# Handling Time Series & Combining
df['Date'] = pd.to_datetime('2024-' + df['Month'] + '-01', errors='coerce')
print("\nData with Time Series Date:\n", df[['Date', 'Sales']].head())

IDLE Shell 3.14.4
File Edit Shell Debug Options Window Help
Sequence: [ 0. 2.5 5. 7.5 10. ]
Standard Deviation: 1.118033988749895
Bitwise AND: [1 0]
Broadcasting: [[11 12]
[13 14]]
Sorted: [1 2 5 8]
Non-zero count: 4
>>>
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Practical_04.py
Pandas Series:
0 10
1 20
2 30
dtype: int64
Displaying 10 Grains of Data:
Month Product City Sales Quantity
0 Jan Laptop Mumbai 50000 1
1 Feb Mouse Delhi 1200 2
2 March Laptop Mumbai 50000 1
3 Jan Phone Pune 20000 1
4 Feb Laptop Mumbai 50000 1
5 March Phone Pune 20000 1
6 Jan Mouse Delhi 1200 2
7 April Laptop Pune 50000 1
8 May Phone Mumbai 20000 1
9 June Laptop Delhi 50000 1
Best Month for Sales: Jan
Product Sold Most (Quantity): Laptop
City with Most Products: Delhi
Data with Time Series Date:
Date Sales
0 2024-01-01 50000
1 2024-02-01 1200
2 NaT 50000
3 2024-01-01 20000
4 2024-02-01 50000
Ln: 173 Col: 0
```

MITAOE Home Dashboard My courses

Not secure moodle.mitaoe.ac.in/course/view.php?id=6094

Python Basics.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

cricket.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

Student Marks tuples.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/A...

```
# Marks in a list
marks = [85, 90, 78, 92, 88]
total = sum(marks)
percentage = total / len(marks)

print(f"Total Marks: {total}, Percentage: {percentage}%")

# Tuple Operations
my_tuple = (1, 2, 3, 2, 4)
print("Tuple Count of 2:", my_tuple.count(2))
print("Index of 3:", my_tuple.index(3))
```

File Edit Format Run Options Window Help

File Edit Shell Debug Options Window Help

```
Enter a number: 222
Cube Root = 6.055048946511104

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Single Digit -> Square, Two Digit -> Square Root, Three Digit -> Cube Root.py
Enter a number:
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/statistical analysis.py
Enter number of employees: 55000
Enter salary of employee 1: 30000
Enter salary of employee 2: 85000
Enter salary of employee 3: 60000
Enter salary of employee 4:
= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/statistical analysis.py
Enter number of employees: 3
Enter salary of employee 1: 20000
Enter salary of employee 2: 50000
Enter salary of employee 3: 60000
Sum = 130000.0
Average = 43333.333333333336
Maximum = 60000.0
Minimum = 20000.0

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Python Basics.py
Enter number to search: 40
Number found at index 3

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/cricket.py
The Captain of the team is: Hardik (Height: 183cm)

= RESTART: C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi, Pune/Attachments/ctlab1_lab2.html/EDS_LAB/Student Marks tuples.py
Total Marks: 433, Percentage: 86.6%
Tuple Count of 2: 2
Index of 3: 2
```

Ln: 105 Col: 0

5:31 AM

30°C Clear

23:00 09-05-2026

MITAOE Home Dashboard My courses

Python Basics.py - C:/Users/prati/OneDrive - MIT Academy of Engineering (MITAOE), Alandi...

```
def linear_search(arr, target):
    for i in range(len(arr)):
        if arr[i] == target:
            return i # Position found
    return -1

numbers = [10, 20, 30, 40, 50]
target = int(input("Enter number to search: "))
pos = linear_search(numbers, target)

if pos != -1:
    print(f"Number found at index {pos}")
else:
    print("Number not found.")
```

Ln: 15 Col: 0

Pre Course survey

Opened: Saturday, 18 April 2026, 5:31 AM Closes: Saturday, 30 May 2026, 5:31 AM

30°C Clear

Search

ENG IN

22:56 09-05-2026

File Edit Shell Debug Options Window Help

untitled (3,14,4)

```
>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit - Square, Two Digit - S
quare Root, Three Digit - Cube Root.py
Enter a number: 55
Square Root = 7.416198487095663

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit - Square, Two Digit - S
quare Root, Three Digit - Cube Root.py
Enter a number: 222
Cube Root = 6.055048946511104

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Single Digit - Square, Two Digit - S
quare Root, Three Digit - Cube Root.py
Enter a number:
= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\statistical analysis.py
Enter number of employees: 55000
Enter salary of employee 1: 30000
Enter salary of employee 2: 65000
Enter salary of employee 3: 60000
Enter salary of employee 4:
= RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\statistical analysis.py
Enter number of employees: 3
Enter salary of employee 1: 20000
Enter salary of employee 2: 50000
Enter salary of employee 3: 60000
Sum = 130000.0
Average = 43333.333333333336
Maximum = 60000.0
Minimum = 20000.0

>>> = RESTART: C:\Users\prati\OneDrive - MIT Academy of Engineering (MITAOE), Alandi
, Pune\Attachments\ctlab1_lab2.html\EDS_LAB\Python Basics.py
Enter number to search: 40
Number found at index 3

>>>
```

Ln: 97 Col: 0